

14..py - C:/Users/Admin/Downloads/14..py (3.13.13)

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```
import matplotlib.pyplot as plt
import numpy as np

# Student study time (hours) and exam scores
study_time = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
exam_score = [45, 50, 55, 60, 65, 70, 75, 80, 88, 92]

# Calculate correlation coefficient
correlation = np.corrcoef(study_time, exam_score)[0, 1]

print("Study Time and Exam Score Analysis")
print("-----")
print("Correlation coefficient =", round(correlation, 2))

if correlation > 0:
    print("Result: There is a positive correlation.")
elif correlation < 0:
    print("Result: There is a negative correlation.")
else:
    print("Result: There is no correlation.")

# 1. Scatter Plot
plt.figure()
plt.scatter(study_time, exam_score)
plt.title("Study Time vs Exam Score")
plt.xlabel("Study Time (Hours)")
plt.ylabel("Exam Score")
plt.grid(True)
plt.show()

# 2. Line Plot
plt.figure()
plt.plot(study_time, exam_score, marker='o')
plt.title("Study Time vs Exam Score - Line Plot")
plt.xlabel("Study Time (Hours)")
plt.ylabel("Exam Score")
plt.grid(True)
plt.show()
```

```
# 3. Bar Plot
plt.figure()
plt.bar(study_time, exam_score)
plt.title("Study Time vs Exam Score - Bar Plot")
plt.xlabel("Study Time (Hours)")
plt.ylabel("Exam Score")
plt.show()
```

```
IDLE Shell 3.13.13
File Edit Shell Debug Options Window Help
Python 3.13.13 (tags/v3.13.13:01104ce, Apr 7 2026, 19:25:48) [MSC v.1944 64 bit
(AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/Admin/Downloads/14..py =====
Study Time and Exam Score Analysis
-----
Correlation coefficient = 1.0
Result: There is a positive correlation.
>>>
```



Figure 1



Study Time vs Exam Score

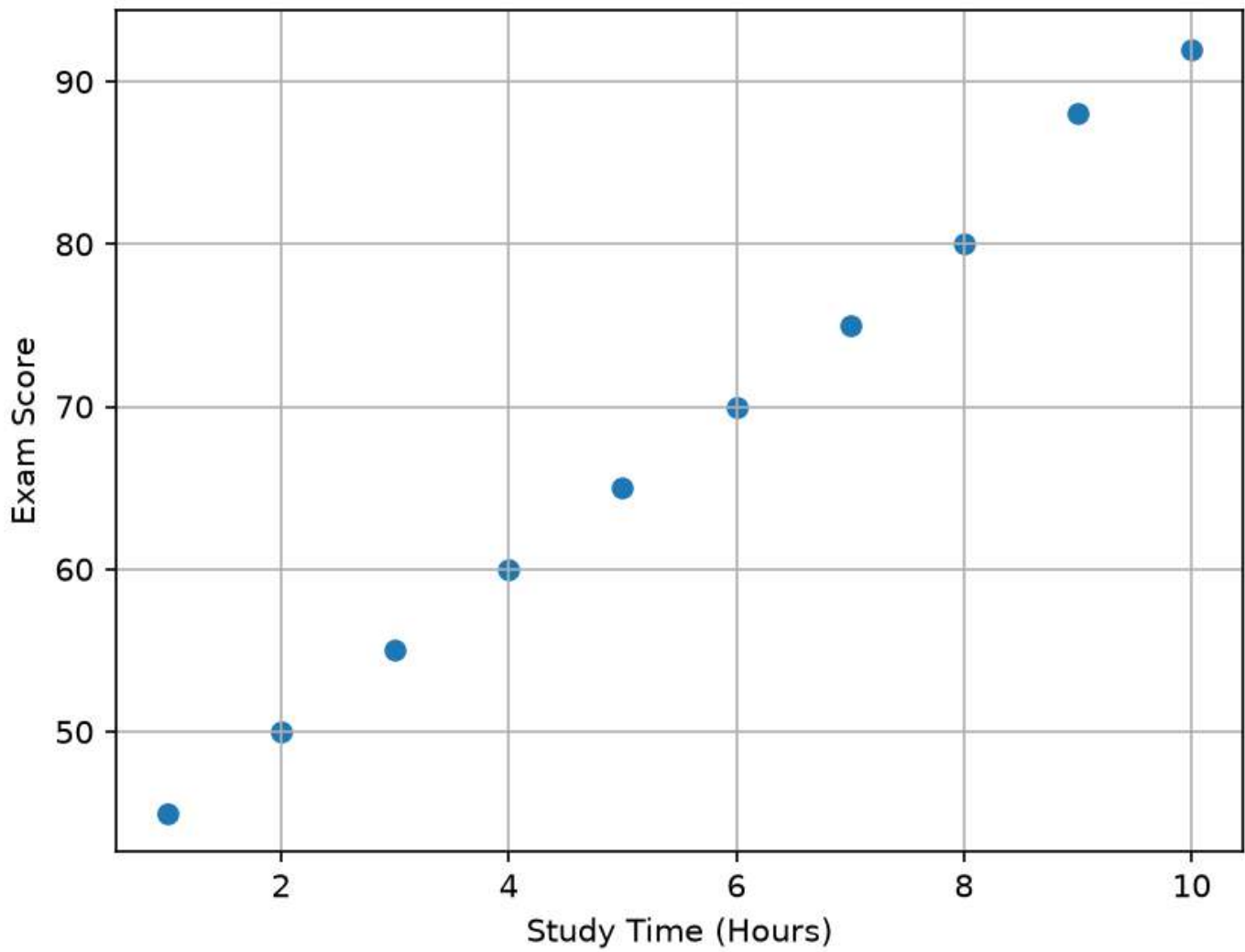
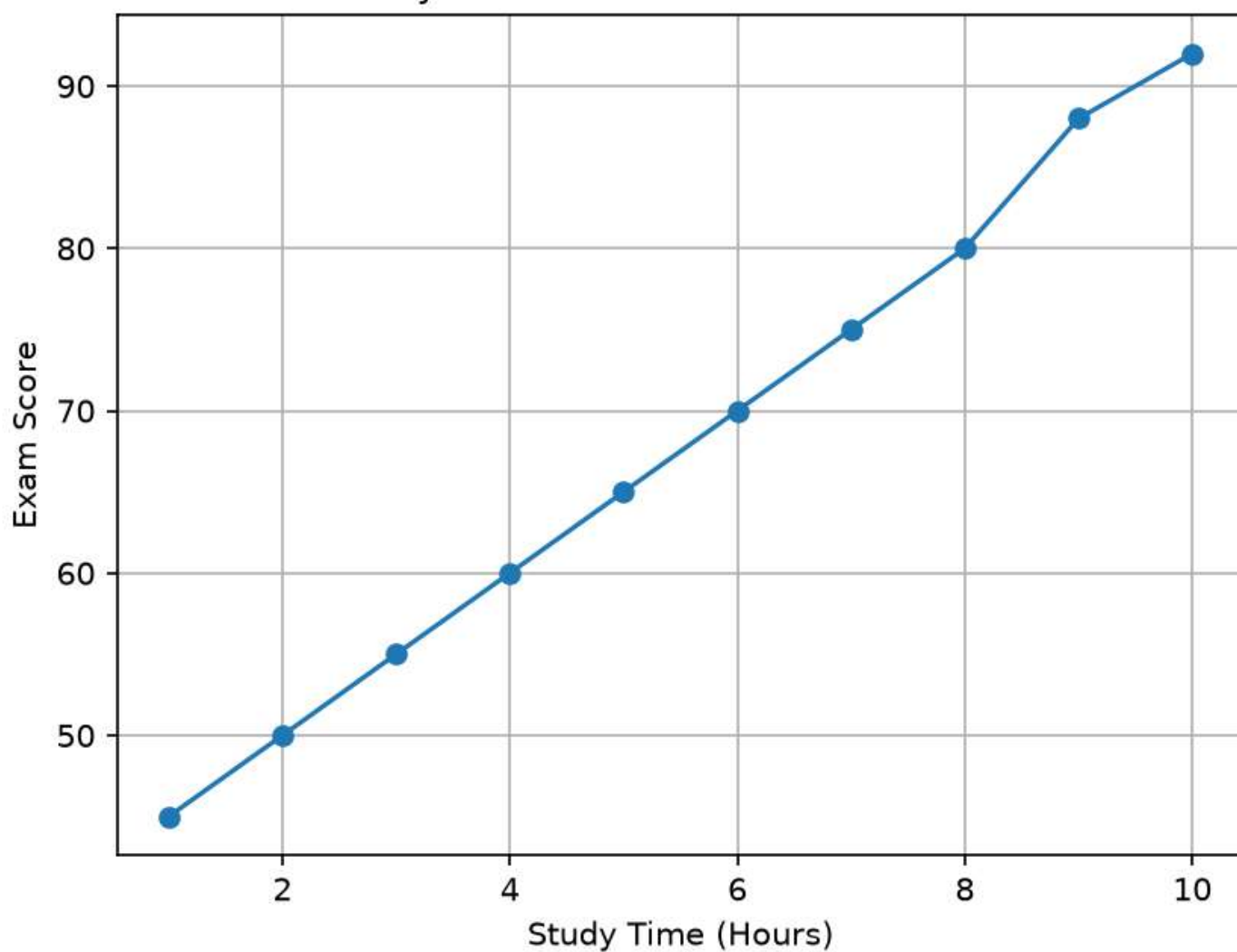




Figure 1



Study Time vs Exam Score - Line Plot



Study Time vs Exam Score - Bar Plot

